

Project 2 Report: Regression Analysis of Malibu-Area Housing Prices

Course: AMS 315

Author: Ernest Nagapetov

1. Introduction and Research Question

Research question: Which property characteristics best predict home prices?

The objective of this project is to perform a multiple linear regression analysis on housing data from the Malibu area. By analyzing the provided structural, geographic, and community features, the aim is to construct a predictive model while addressing statistical violations that occur in the provided real estate data, which may be significant.

2. Data Overview

The dataset contains information on $n = 3,000$ residential properties with 15 price predictor features.

Important Stats of the Target Variable:

- **Mean Price:** \$2,556,325
- **Standard Deviation:** \$2,625,727
- **Minimum Price:** \$250,000
- **Median (50%) Price:** \$1,740,820
- **Maximum Price:** \$11,669,030

The significant difference between the median and maximum values and the high standard deviation indicate a right-skewed distribution, which is typical of luxury real estate markets.

3. Correlation Analysis

A correlation analysis was executed to evaluate the linear relationships between the predictors and the target variable.

Correlation Matrix Results

- **Strongest Positive Predictor:** house_size shares a strong linear relationship with price ($r = 0.94$). Other size metrics like number_of_rooms ($r = 0.85$) and number_of_bathrooms ($r = 0.77$) also have significant positive correlations.
- **Strongest Negative Predictor:** distance_to_malibu has a significant negative correlation ($r = -0.71$). This shows that properties closer to the core Malibu area yield higher premiums.

- **Properties near Beach:** distance_to_the_beach displays a negative correlation ($r = -0.29$), showing that closer proximity to the ocean increases value.
- **Possibility of Multicollinearity:** High intercorrelations are evident among size features (as seen in house_size vs. number_of_rooms, which has $r = 0.91$), meaning that there is a high likelihood of redundancy.

4. Original Model and Improvements

Step 1: Original Model

The initial approach used an Ordinary Least Squares (OLS) model with three major property value drivers: house_size, size_of_the_yard, and distance_to_the_beach.

Formula: $y_{\text{price}} = \beta_0 + \beta_1(\text{house_size}) + \beta_2(\text{size_of_the_yard}) + \beta_3(\text{distance_to_the_beach}) + \epsilon$

- **R² Score:** 0.9591
- **RMSE:** \$513,732.13
- **Estimated Coefficients:**
 - Intercept (β_0) = -\$1,455,149
 - house_size (β_1) = +\$1,537.26 per sq. ft.
 - size_of_the_yard (β_2) = +\$49.98 per sq. ft.
 - distance_to_the_beach (β_3) = -\$377,905.40 per mile

All three features were highly statistically significant ($p \approx 0.0$). This model uses only 3 parameters to attempt to explain the variance.

Step 2: Feature Selection via Stepwise Forward Selection (AIC)

To find out if other structural variables could optimize predictive performance, a forward stepwise selection strategy based on the Akaike Information Criterion (AIC) was used. The reason AIC is used is that it penalizes unnecessary complexity and rewards improvements in goodness-of-fit.

1. Started with house_size alone (AIC = 90,768)
2. The inclusion of size_of_the_yard and distance_to_the_beach yielded the largest initial drops in AIC.
3. The algorithm iteratively added variables, stopping when additional features no longer minimized the criterion. 13 of the 15 candidate features were selected, with solar_panel and distance_to_santa_monica being excluded.
4. Final AIC = 86,223.

Step 3: Final Model

The model utilizing all 13 AIC-selected features yielded the following results:

- **Full Model R² / Adjusted R²:** 0.9672 / 0.9671

- **Overall F-Statistic:** 29,454 ($p \approx 0.0$), indicating major global significance.

Moving from the simple 3-feature model to the expanded 13-feature model added ten parameters but only improved the R^2 value by 0.8%, indicating diminishing returns.

5. Model Diagnostics

Even though both models offer good predictive accuracy, standard OLS linear inference depends heavily on unviolated assumptions. Checking model diagnostics shows the structural failures that must be resolved.

1. Multicollinearity (Variance Inflation Factor)

The Variance Inflation Factor (VIF) measures how severely a feature's variance is inflated by cross-correlations with other independent variables.

Feature	VIF Value	Collinearity Status (Rule of thumb: $VIF < 5$ = fine, > 10 = problem, $>> 10$ = severe)
lot_size	1462.38	severe multicollinearity
size_of_the_yard	1170.87	severe multicollinearity
number_of_rooms	72.38	severe multicollinearity
house_size	66.55	severe multicollinearity
number_of_bathrooms	56.31	severe multicollinearity
school_district	49.40	severe multicollinearity

number_of_parking_spot	29.44	severe multicollinearity
distance_to_malibu	21.04	severe multicollinearity

- **Result:** lot_size, yard, rooms, and house_size are redundant (all measure size one way or another)
- **Consequence:** Such high VIF scores cause the OLS regression coefficients to become volatile, even though the model yields a high overall R^2 value.

2. Heteroskedasticity (Breusch-Pagan Test & Residual Plot)

OLS assumes homoskedasticity (constant error variance across all predicted levels).

- **Breusch-Pagan Test:** LM Statistic = 212.88, $p = 0.000000$. We reject H_0 (null hypothesis of constant variance).
- **Residual Plot Analysis:** The residual plot takes the shape of a funnel, meaning that the vertical spread of the residuals widens significantly as the predicted price grows. Additionally, a sharp linear segment near zero shows that the model struggles with lower-value properties, signifying a non-linear relationship.
- **Consequence:** The calculated standard errors are biased, meaning that the standard OLS t-statistics and p-values are unreliable for hypothesis testing.

3. Normality of Errors (Q-Q Plot Analysis)

The Normal Quantile-Quantile (Q-Q) plot validates whether the error residuals follow a normal distribution.

- **Observations:** The residuals track the theoretical normal reference line closely through the middle range but deviate significantly at both extreme tails. The upper tail pulls sharply away and upward.
- **Interpretation:** The distribution exhibits leptokurtic behavior, meaning that it has a taller, sharper peak and a bigger tail than a standard bell-curve. This is due to the luxury high-value real estate outliers in the Malibu area, inflating the model's upper errors.

6. Conclusions, Limitations, and Fixes

Conclusions:

- Model explains 96.7% of price variance, indicating strong predictive performance
- House size and proximity to Malibu are the dominant price drivers
- The 3-feature model nearly matches the 13-feature model (0.8% difference in R^2)
- RMSE of ~\$514K is reasonable as a baseline for luxury real estate

Limitations:

- Severe multicollinearity (lot_size VIF = 1,462), meaning individual coefficients can't be interpreted
- Heteroskedasticity was confirmed, meaning standard errors and p-values are unreliable without correction

Fixes:

- Model log(price) instead of raw price to stabilize variance
- Apply robust standard errors (HC3) to correct for heteroskedasticity
- Consolidate or drop redundant size features to address collinearity

7. Final Model

Comparison:

Metrics	First Model	Final Model
Predictors	3	14
Response Variable	Price	log(Price)
R ²	0.967	0.855
Adjusted R ²	0.967	0.854
RMSE	\$457,108	\$1,656,765
Heteroscedasticity Correction	None	HC3 Robust SE
F-statistic	29,454	1,172
BP LM Statistic	212.88	295.92
BP F-Statistic	57.19	23.96
BP p-value	≈ 0.000	≈ 0.000

Note: The two R² values are not directly comparable since the value in the final model is on a log scale.

Difference between the two models:

The main difference between the two models is shown in the BP Test, which confirmed significant heteroscedasticity in the first model (LM = 212.88, $p \approx 0.000$), meaning its standard errors and p-values were unreliable. The final model was done by applying robust standard errors (HC3) to correct for heteroskedasticity. What HC3 Robust SE does is it adjusts the standard errors so that the p-values, test statistics, and confidence intervals remain accurate. After running the final model, the F-statistic dropped from 57.19 to 23.96, showing the log transformation caused a significant reduction in the severity of heteroscedasticity. The final model also produced a Durbin-Watson statistic of 1.933, which is very close to the ideal value of 2 and close to the value for the first model (2.050), meaning that the final model didn't sacrifice autocorrelation in order to fix heteroscedasticity.

Conclusions of the Final Model:

What Drives Price Up

- Number of rooms ($\beta = +0.1950$)
- Number of parking spots ($\beta = +0.1114$)
- Solar panels ($\beta = +0.1079$)
- HOA ($\beta = +0.0894$)
- Gated ($\beta = +0.0786$)
- Number of bathrooms ($\beta = +0.0515$)

What Drives Price Down

- Distance to the beach ($\beta = -0.2595$)
- School district ($\beta = -0.2109$)
- Distance to Malibu ($\beta = -0.0137$)
- Distance to Santa Monica ($\beta = -0.0053$)

